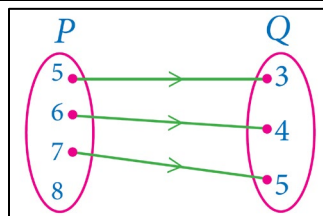


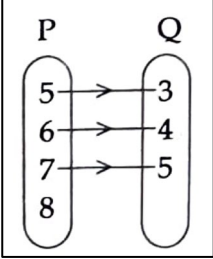
10 TH MATHS PUBLIC EXAM QUESTIONS-2026-2027

CHAPTER 1: RELATIONS AND FUNCTIONS

2 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	If $A \times B = \{(3,2), (3,4), (5,2), (5,4)\}$ then find A and B.	Sep 2019, Aug 2022
2	Show that the function $f: N \rightarrow N$ defined by $f(m) = m^2 + m + 3$ is one-one function.	Sep 2019
3	If $A = \{1,3,5\}$ and $B = \{2,3\}$ then find. Show that $n(A \times B) = n(B \times A) = n(A) \times n(B)$	Sep 2021
4	Let $A = \{1,2,3,4, \dots 45\}$ and R Be the relation defined as "is square of on A". write R as a subset of $A \times A$. Also, find the domain and range of R.	Sep 2021
5	Let $A = \{1,2,3\}$ and $B = \{x x \text{ is a prime number less than } 10\}$. Find $A \times B$ and $B \times A$	May 2022
6	The arrow diagram shows in figure a relationship between the sets P and Q. Write the relation in (i) Set builder form (ii) Roster form (iii) What is the domain and range of R.	May 2022
7	If $A = \{5,6\}$, $B = \{4,5,6\}$, $C = \{5,6,7\}$. Show that $A \times A = (B \times B) \cap (C \times C)$.	Aug 2022
8	If $B \times A = \{(-2,3), (-2,4), (0,3), (0,4), (3,3), (3,4)\}$ find A and B	April 2023 July 2026
9	Find 'k' if $f \circ f(k) = 5$ where $f(k) = 2k - 1$.	April 2023
10	A Relation R is given by the set $\{(x,y)/y = x + 3, x \in \{0,1,2,3,4,5\}\}$. Determine its domain and the range.	June 2023, April 2026
11	Check whether $(f \circ g) = (g \circ f)$ if $f(x) = x - 6$ and $g(x) = x^2$	June 2023
12	If $A \times B = \{(3,2), (3,4), (5,2), (5,4)\}$ then find A and B.	April 2024
13	If $f(x) = 3x - 2$, $g(x) = 2x + k$ and $f \circ g = g \circ f$, then find value of k.	April 2024
14	If $B \times A = \{(-2,3), (-2,4), (0,3), (0,4), (3,3), (3,4)\}$ find A and B.	July 2024
15	Given $f(x) = 2x - x^2$. Find (i) $f(1)$ (ii) $f(x + 1)$.	July 2024
16	Let $A = \{1,2,3\}$ and $B = \{x x \text{ is a prime number less than } 10\}$. Find $A \times B$	April 2025
17	Find $f \circ g$ and $g \circ f$ when $f(x) = 2x + 1$ and $g(x) = x^2 - 2$.	April 2025
18	If $A = \{2, -2, 3\}$, $B = \{1, -4\}$ then find $A \times B$, $B \times A$.	July 2025



S.No	PUBLIC QUESTIONS	YEAR
19	The arrow diagram shows the relationship between the sets P and Q. Write the relation in (i) Set builder form (ii) Roster form 	July 2025
20	Let $X = \{-5, 1, 3, 4\}$ and $Y = \{a, b, c\}$ Determine whether the relation $R = \{(-5, a), (1, a), (3, b)\}$ is a function from X to Y?	April 2026
21	If $f(x) = x^2 - 1$, $g(x) = x - 2$ find a, if $(g \circ f)(a) = 1$	July 2026
22	A relation R is given by the set $R = \{(x, y)/y = x^2 + 3, x \in \{0, 1, 2, 3, 4, 5\}\}$. Determine its domain and range.	(PTA-2)
23	Let $A = \{1, 2, 3, \dots, 100\}$ and R be the relation defined as “is cube of” on A. Find the domain and range of R.	(PTA-4)
24	Let $A = \{1, 2, 3, 4\}$ and $B = \mathbb{N}$. Let $f: A \rightarrow B$ be defined by $f(x) = x^2$, find (i) the range of f (ii) identify the type of function.	(PTA-5)
25	If $R = \{(x, -2), (-5, y)\}$ represents the identity functions, find the values of x and y.	(PTA-6)

5 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Let $A =$ The set of all natural numbers less than 8, $B =$ The set of all prime numbers less than 8, $C =$ The set of even prime number. Verify $(A \cap B) \times C = (A \times C) \cap (B \times C)$	Sep 2019, July 2024
2	Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 5, 8, 11, 14\}$ be two sets. Let $f: A \rightarrow B$ be a function give by $f(x) = 3x - 1$. Represent this function (i) by arrow diagram (ii) in a table form (iii) as a set of ordered pairs (iv) in a graphical form	Sep 2019, July 2025
3	Let $A = \{0, 1, 2, 3\}$ and $B = \{1, 3, 5, 7, 9\}$ be two sets. Let $f: A \rightarrow B$ be a function give by $f(x) = 2x - 1$. Represent this function (i) by arrow diagram (ii) in a table form (iii) as a set of ordered pairs (iv) in a graphical form	Sep 2019, April 2024
4	Let $A = \{x \in W x < 2\}$, $B = \{x \in N 1 < x \leq 4\}$ and $C = \{3, 5\}$. Verify that (i) $A \times (B \cap C) = (A \times B) \cap (A \times C)$ (ii) $A \times (B \cup C) = (A \times B) \cup (A \times C)$	Sep 2021, April 2026, July 2026
5	Let $A =$ The set of all natural numbers less than 8, $B =$ The set of all prime numbers less than 8, $C =$ The set of even prime number. Verify $A \times (B - C) = (A \times B) - (A \times C)$	May 2022
6	Represent each of the given relations by (a) an arrow diagram, (b) a graph and (c) a set in roster form, wherever possible. $\{(x, y) y = x + 3, x, y \text{ are natural numbers } < 10\}$	Aug 2022

S.No	PUBLIC QUESTIONS	YEAR
7	Let $f: A \rightarrow B$ be a function defined by $f(x) = \frac{x}{2} - 1$, where $A = \{2, 4, 6, 10, 12\}$, $B = \{0, 1, 2, 4, 5, 9\}$. Represent f by (i) Set of ordered pairs, (ii) a table (iii) a graph (iv) an arrow diagram	April 2023, April 2025
8	Let $A = \{x \in W x < 3\}$, $B = \{x \in N 1 < x \leq 5\}$ and $C = \{3, 5, 7\}$. Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$	April 2023
9	Let $A = \{x \in W x < 2\}$, $B = \{x \in N 1 < x \leq 4\}$ and $C = \{3, 5\}$. Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$	Sep 2023
10	Let $A = \{x \in N 1 < x < 4\}$, $B = \{x \in W 0 \leq x < 2\}$ and $C = \{x \in W x < 3\}$. Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$	April 2024
11	A function $f: [-5, 9] \rightarrow \mathbb{R}$ is defined as follows: $f(x) = \begin{cases} 6x + 1; & -5 \leq x < 2 \\ 5x^2 - 1; & 2 \leq x < 6 \\ 3x - 4; & 6 \leq x \leq 9 \end{cases}$ Find (i) $f(-3) + f(2)$ (ii) $f(7) - f(1)$ (iii) $2f(4) + f(8)$ (iv) $\frac{2f(-2) - f(6)}{f(4) + f(-2)}$	July 2024
12	Let $A =$ The set of all natural numbers less than 8, $B =$ The set of all prime numbers less than 8, $C =$ The set of even prime numbers. Verify that $(A \cap B) \times C = (A \times C) \cap (B \times C)$	April 2025
13	If $f(x) = x - 1$, $g(x) = 3x + 1$ and $h(x) = x^2$, then show that $(f \circ g) \circ h = f \circ (g \circ h)$	July 2025, April 2026
14	Let f be a function $f: N \rightarrow N$ be defined by $f(x) = 3x + 2$, $x \in N$ (i) Find the images of 1, 2, 3. (ii) Find the pre-images of 29, 53. (iii) Identify the type of function.	July 2026
15	Let $A = \{x \in W / 0 < x < 5\}$, $B = \{x \in W / 0 \leq x \leq 2\}$, $C = \{x \in W / x < 3\}$ then verify that, $A \times (B \cap C) = (A \times B) \cap (A \times C)$.	(PTA-3)

CHAPTER 2: NUMBERS AND SEQUENCES

2 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	If m, n , are natural numbers, for what values of m , does $2^n \times 5^m$ ends in 5?	Sep 2019
2	Find the 3 rd and 4 th terms of a sequence, if $a_n = \begin{cases} n^2 & \text{if } n \text{ is odd} \\ \frac{n^2}{2} & \text{if } n \text{ is even} \end{cases}$	Sep 2019
3	Find the value of $1^2 + 2^2 + 3^2 + \dots + 10^2$ and hence deduce $2^2 + 4^2 + 6^2 + \dots + 20^2$	Sep 2019
4	Find the number of terms in the A.P. 3, 6, 9, 12, ... 111.	Sep 2021

S.No	PUBLIC QUESTIONS	YEAR
5	If $3 + k$, $18 - k$, $5k + 1$ are in A.P. then find the value of ' k '.	Sep 2021
6	Which term of an A.P. 16, 11, 6, 1, ... is -54 ?	May 2022, July 2024
7	If $13824 = 2^a \times 3^b$ then find ' a ' and ' b '	May 2022
8	Find the least number that is divisible by the first ten natural numbers.	Aug 2022, June 2023
9	Find the 19^{th} term of an A.P. $-11, -15, -19, \dots$	Aug 2022
10	Find x so that $x + 6$, $x + 12$ and $x + 15$ are consecutive terms of a Geometric Progression.	April 2023
11	If $p^2 \times q^1 \times r^4 \times s^3 = 3,15,000$ then find the values of ' p ', ' q ', ' r ' and ' s '	April 2023
12	Find the 8^{th} term of the G.P. 9, 3, 1,	June 2023, July 2024, Apr 2026
13	' a ' and ' b ' are two positive integers such that $a^b \times b^a = 800$. Find ' a ' and ' b '	April 2024, July 2025
14	Find the HCF of 23 and 12.	April 2024
15	A man has 532 flower pots. He wants to arrange them in rows such that each row contains 21 flower pots. Find the number of completed rows and how many flower pots are left over.	April 2025
16	Check whether the sequence $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \sqrt{50}, \dots$ is an A.P or not?. If it is an A.P, then find the common difference.	April 2025
17	In a G.P. 729, 243, 81 ... find t_7	July 2025
18	Find the least positive value of x such that $71 \equiv x \pmod{8}$	April 2026
19	Find the terms a_6 and a_{13} of the sequence whose n^{th} term is given by $a_n = \frac{5n}{n+2}$	July 2026
20	Find the sum of: $1 + 3 + 5 + \dots + 55$	July 2026
21	If $1^3 + 2^3 + 3^3 + \dots + k^3 = 16900$, then find $1 + 2 + 3 + \dots + k$.	(PTA-3)
22	An organisation plans to plant saplings in 25 streets in a town in such a way that one sapling for the first street, three for the second, nine for the third and so on. How many saplings are needed to complete the work?	(DMQ/PTA-1)
23	In a G.P. $\frac{1}{4}, -\frac{1}{2}, 1, -2, \dots$, find t_{10} .	(PTA-4)
24	Which term of the A.P. 21, 18, 15, ... is -81 ? State with reason is there any term 0 in this A.P.?	(PTA-5)
25	Find the common difference of an A.P. in which $t_{18} - t_{14} = 32$.	(PTA-6)

5 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Find the sum of all natural numbers between 100 and 1000 which are divisible by 11.	Sep 2019
2	The sum of three consecutive terms that are in A.P. is 27 and their product is 288. Find the three terms.	Sep 2021
3	Find the HCF of 396, 504, 636	Sep 2021, July 2025
4	If l^{th} , m^{th} and n^{th} terms of an A.P. are, x, y, z respectively, then show that (i) $x(m - n) + y(n - l) + z(l - m) = 0$ (ii) $(x - y)n + (y - z)l + (z - x)m = 0$	May 2022
5	The ratio of 6^{th} and 8^{th} term of an A.P. is 7: 9. Find the ratio 9^{th} term to 13^{th} term.	May 2022
6	Find the largest number which divides 1230 and 1926 leaving remainder 12 in each case.	Aug 2022
7	In an A.P nine times ninth term is equal to the fifteen times fifteenth term, show that six times twenty fourth term is zero.	Aug 2022
8	The houses of a street are numbered from 1 to 49. Senthil's house is numbered Such that the sum of numbers of the houses prior to Senthil's house is equal to the sum of numbers of the houses following Senthil's house. Find Senthil's house number?.	April 2023
9	Find the sum to n terms of the series $5 + 55 + 555 + \dots$	April 2023
10	Find the sum to n terms of the series $3 + 33 + 333 + \dots$	June 2023, July 2024, Apr 2026
11	Rekha has 15 square Colour papers of sizes 10 cm, 11 cm, 12 cm, ... 24 cm. How much area can be decorated with these Colour papers.	June 2023, July 2024, July 2025
12	Find the Sum of $9^3 + 10^3 + \dots + 21^3$.	April 2024
13	Find the sum to n terms of the series $7 + 77 + 777 + \dots$	April 2024
14	If $p_1^{x_1} \times p_2^{x_2} \times p_3^{x_3} \times p_4^{x_4} = 113400$ where p_1, p_2, p_3, p_4 are primes in ascending order and x_1, x_2, x_3, x_4 are integers, find the value of p_1, p_2, p_3, p_4 and x_1, x_2, x_3, x_4 .	April 2025
15	Find the sum of the series $6^2 + 7^2 + 8^2 + \dots + 21^2$.	April 2025
16	Find the sum of all natural numbers between 300 and 600 which are divisible by 7.	April 2026
17	A mother divides into three parts such that the amounts are in A.P. and gives it to her three children. The product of the two least amounts that the children had is 4,623. Find the amount received by each child.	July 2026

S.No	PUBLIC QUESTIONS	YEAR
18	Find the sum of the Geometric series: $3 + 6 + 12 + \dots + 1536$	July 2026
19	Find the sum of all 3 digit natural numbers which are divisible by 9.	(PTA-3)
20	If the sum of the first p terms of an A.P. is $ap^2 + bp$. Find the common difference.	(PTA-6)

CHAPTER 3: ALGEBRA

2 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Find the value of k for which the equation $9x^2 + 3kx + 4 = 0$ has real and equal roots.	Sep 2019
2	If $A = \begin{pmatrix} \sqrt{7} & -3 \\ -\sqrt{5} & 2 \\ \sqrt{3} & -5 \end{pmatrix}^T$ then find the transpose of $-A$.	Sep 2019
3	Determine the quadratic equation whose sum and product of roots are -9 and 20 .	Sep 2021 July 2026
4	Determine the nature of the roots for the quadratic equation $15x^2 + 11x + 2 = 0$	Sep 2021, June 2023, July 2024
5	If $P = \frac{x}{x+y}$, $Q = \frac{y}{x+y}$ then find $\frac{1}{(P^2-Q^2)}$	May 2022
6	Find the excluded values of the following expression $\frac{7p+2}{8p^2+13p+5}$	May 2022
7	Find the value of 'x', in $x^2 - 4x - 12 = 0$	Aug 2022
8	Find the square root of the following rational expressions. $\frac{400x^4y^{12}z^{16}}{100x^8y^4z^4}$	Aug 2022
9	Simplify: $\frac{x+2}{4y} \div \frac{x^2-x-6}{12y^2}$.	April 2023
10	Determine the nature of roots for the following quadratic equation. $2x^2 - x - 1 = 0$	April 2023
11	If $A = \begin{pmatrix} 5 & 2 & 2 \\ -\sqrt{17} & 0.7 & \frac{5}{2} \\ 8 & 3 & 1 \end{pmatrix}$, then verify $(A^T)^T = A$	June 2023
12	Simplify: $\frac{4x^2y}{2z^2} \times \frac{6xz^3}{20y^4}$	April 2024
13	Find the sum and product of the roots for following quadratic equation $x^2 + 8x - 65 = 0$.	April 2024
14	Find the LCM of $9a^3b^2$ and $12a^2b^2c$.	July 2024
15	Add: $\frac{x^3}{x-y} + \frac{y^3}{y-x}$	April 2025

S.No	PUBLIC QUESTIONS	YEAR
16	If $A = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}, B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ find AB and BA. Verify if $AB = BA$.	April 2025
17	Find the excluded values of the expression $\frac{t}{t^2-5t+6}$, if any.	July 2025
18	Given that $A = \begin{pmatrix} -1 & 1 \\ a & b \end{pmatrix}$ where $a, b \geq 0$, find the values of a and b if $A^2 = I$.	July 2025
19	Reduce the rational expression $\frac{x^2-16}{x^2+8x+16}$ to its lowest form.	April 2026
20	If the difference between a number and its reciprocal is $\frac{24}{5}$. find the number.	April 2026
21	Construct a 3×3 matrix whose elements are given by $a_{ij} = i - 2j $.	April 2026
22	Simplify: $\frac{x^2-16}{x+4} \div \frac{x-4}{x+4}$	July 2026
23	Verify that $A^2 = I$ when $A = \begin{pmatrix} 5 & -4 \\ 6 & -5 \end{pmatrix}$	July 2026
24	If two positive integers p and q are written as $p = a^2b^3$ and $q = a^3b$; a, b are prime numbers, then verify $\text{LCM}(p, q) \times \text{HCF}(p, q) = pq$.	(PTA - 2)
25	Find the sum and product of the roots of the equation $8x^2 - 25 = 0$.	(PTA-4)
26	Find the value of p, when $px^2 + (\sqrt{3} - \sqrt{2})x - 1 = 0$ and $x = \frac{1}{\sqrt{3}}$ is one root of the equations.	(PTA-5)

5 Mark Questions

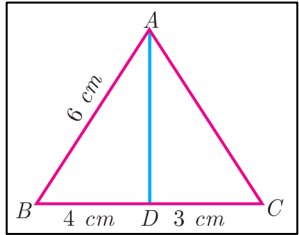
S.No	PUBLIC QUESTIONS	YEAR
1	Solve: $6x + 2y - 5z = 13, 3x + 3y - 2z = 13, 7x + 5y - 3z = 26$	Sep 2019
2	Find the GCD of the given polynomials $x^4 + 3x^3 - x - 3$ and $x^3 + x^2 - 5x + 3$	Sep 2019
3	Find the square root of the following polynomials by division method. $\frac{x^2}{y^2} - \frac{10x}{y} + 27 - \frac{10y}{x} + \frac{y^2}{x^2}$	Sep 2019
4	If $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ -1 & 4 \\ 0 & 2 \end{pmatrix}$ then show that $(AB)^T = B^T A^T$	Sep 2021, April 2023, April 2025, April 2026
5	If the equation $(1 + m^2)x^2 + 2mcx + c^2 - a^2 = 0$ has equal roots, then prove that $c^2 = a^2(1 + m^2)$	Sep 2021
6	Solve $x^2 + 2x - 2 = 0$ by Formula method.	May 2022
7	If $36x^4 - 60x^3 + 61x^2 - mx + n$ is a perfect square, find the values of 'm' and 'n'.	May 2022
8	Solve $pqx^2 - (p + q)^2x + (p + q)^2 = 0$	May 2022

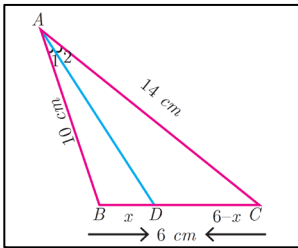
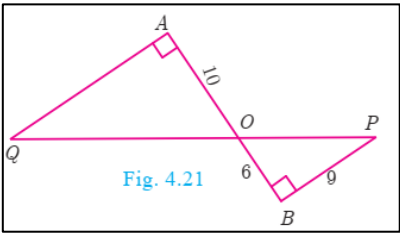
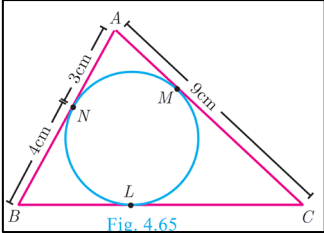
S.No	PUBLIC QUESTIONS	YEAR
9	If α and β are the roots of $7x^2 + ax + 2 = 0$ and $\beta - \alpha = \frac{-13}{7}$, Find the values of a.	May 2022
10	Solve $x + y + z = 5$, $2x - y + z = 9$, $x - 2y + 3z = 16$	Aug 2022
11	Find the square root of $64x^2 - 16x^3 + 17x^2 - 2x + 1 = 0$	Aug 2022, April 2024
12	Simplify: $\frac{b^2+3b-28}{b^2+4b+4} \div \frac{b^2-49}{b^2-5b-14}$	Aug 2022
13	Find the square root of $x^4 - 12x^3 + 42x^2 - 36x + 9$	Aug 2022
14	Solve: $x + 20 = \frac{3y}{2} + 10 = 2z + 5 = 110 - (y + z)$	April 2023
15	Solve $3x - 2y + z = 2$, $2x + 3y - z = 5$, $x + y + z = 6$.	June 2023
16	Find the square root of the following polynomials by division method: $121x^4 - 198x^3 - 183x^2 + 216x + 144$	June 2023
17	If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ show that $A^2 - 5A + 7I_2 = 0$	June 2023, April 2024, July 2025
18	If $9x^4 + 12x^3 + 28x^2 + ax + b$ is a perfect square, find the values of 'a' and 'b'	July 2024
19	If $A = \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$ then show that $(AB)^T = B^T A^T$	July 2024
20	Determine the nature of the roots of the equation $(x - 1)^2 + (x - 2)^2 + (x - 3)^2 = 0$	April 2025
21	Find the values of m and n of the polynomial $36x^4 - 60x^3 + 61x^2 - mx + n$ are perfect squares.	July 2025
22	If α and β are the roots of the equation $2x^2 - 3x + 1 = 0$, then form a quadratic equation whose roots are $\frac{1}{\alpha^2}$ and $\frac{1}{\beta^2}$.	April 2026
23	Find the square root of the polynomial $37x^2 - 28x^3 + 4x^4 + 42x + 9$ by division method.	July 2026
24	Find the matrix X and Y if $X + Y = \begin{bmatrix} 7 & 0 \\ 3 & 5 \end{bmatrix}$ and $X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$	July 2026
25	Find two consecutive positive integers, sum of whose squares is 365.	(DMQ)
26	A motor boat whose speed is 18km/hr in still water takes 1 hour more to go to 24 km upstream than to return downstream to the same spot. Find the speed of the stream.	(DMQ/PTA-2)
27	The base of a triangle is 4cm longer than its altitude. If the area of the triangle is 48 sq.cm, then find its base and altitude.	(PTA-1)
28	Solve: $2x + y + 4z = 15$, $x - 2y + 3z = 13$, $3x + y - z = 2$.	(PTA - 1)

S.No	PUBLIC QUESTIONS	YEAR
29	Find the values of a and b if $16x^4 - 24x^3 + (a - 1)x^2 + (b + 1)x + 49$ is a perfect square.	(PTA-2)
30	Solve the equation $\frac{1}{x+1} + \frac{2}{x+2} = \frac{4}{x+4}$ where $x + 1 \neq 0$, $x + 2 \neq 0$ and $x + 4 \neq 0$ using quadratic formula.	(PTA-3)
31	Simplify: $\frac{a^2-16}{a^3-8} \times \frac{2a^2-3a-2}{2a^2+9a+4} \div \frac{3a^2-11a-4}{a^2-2a+4}$.	(PTA-4)
32	If $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$ show that $A^2 - 4A + 5I_2 = 0$.	(PTA-5)
33	A train covered a certain distance at a uniform speed. If the train would have been 10 km/hr faster it would have taken 2 hour less than the scheduled time and if the train were slower by 10 km/hr, it would have taken 3 hour more than the scheduled time. Find the distance covered by the train.	(PTA-5)
34	$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 3 \\ -1 & 5 \end{pmatrix}$, $C = \begin{pmatrix} -1 & 5 \\ 1 & 3 \end{pmatrix}$ prove that, $A(BC) = (AB)C$.	(PTA-6)
35	A car left 30 minutes later than the scheduled time. In order to reach its destination 150km away in time, it has to increase its speed by 25 km/hr from its usual speed. Find its usual speed.	(PTA-6)

CHAPTER 4: GEOMETRY

2 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Check whether AD is bisector of $\angle A$ of $\triangle ABC$ in each of the following $AB = 5$ cm, $AC = 10$ cm, $BD = 1.5$ cm and $CD = 3.5$ cm	Sep 2019
2	In $\triangle ABC$, D and E are points on the sides AB and AC respectively such that $DE \parallel BC$. If $\frac{AD}{DB} = \frac{3}{4}$ and $AC = 15$ cm find AE.	Sep 2021
3	In the figure AD is the bisector of $\angle A$. If $BD = 4$ cm, $DC = 3$ cm and $AB = 6$ cm, find AC. 	May 2022, April 2025
4	ABCD is a trapezium in which $AB \parallel DC$ and P, Q are points on AD and BC respectively, such that $PQ \parallel DC$. If $PD = 18$ cm, $BQ = 35$ cm and $QC = 15$ cm, find AD.	Aug 2022

S.No	PUBLIC QUESTIONS	YEAR
5	In the figure AD is the bisector of $\angle BAC$, if $AB = 10$ cm, $AC = 14$ cm and $BC = 6$ cm. Find BD and DC. 	April 2023
6	Check whether AD is bisector of $\angle A$ of $\triangle ABC$ in each of the following $AB = 4$ cm, $AC = 6$ cm, $BD = 1.6$ cm and $CD = 2.4$ cm.	June 2023
7	A man goes 18 m due east and then 24 m due North. Find the distance of his current position from the starting point.	April 2024
8	In figure QA and PB are perpendicular to AB. If $AO = 10$ cm, $BO = 6$ cm and $PB = 9$ cm. Find AQ.  Fig. 4.21	July 2024
9	In the given figure, $\triangle ABC$ is circumscribing a circle. Find the length of BC.  Fig. 4.65	July 2025
10	D and E are respectively the points on the sides AB and AC of a $\triangle ABC$ such that $AB = 5.6$ cm, $AD = 1.4$ cm, $AC = 7.2$ cm and $AE = 1.8$ cm, show that $DE \parallel BC$.	April 2026
11	If $\triangle ABC$ is similar to $\triangle DEF$ such that $BC = 3$ cm, $EF = 4$ cm and area of $\triangle ABC = 54$ cm², find the area of $\triangle DEF$.	July 2026
12	P and Q are points on sides AB and AC respectively of $\triangle ABC$. If $AP = 3$ cm, $PB = 6$ cm, $AQ = 5$ cm and $QC = 10$ cm. Show that $BC = 3PQ$.	(PTA-6)

5 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	State and prove Angle Bisector Theorem.	Sep 2019, Aug 2022, April 2023, April 2025
2	State and prove Pythagoras Theorem.	Sep 2021, June 2023, July 2024, July 2025

S.No	PUBLIC QUESTIONS	YEAR
		July 2026
3	Show that in a triangle, the medians are concurrent.	Sep 2021
4	State and Prove Basic Proportionality Theorem (BPT) or Thales theorem.	May 2022, April 2024
5	An Aeroplane leaves an airport and flies due north at a speed of 1000 km/hr. At the same time, another Aeroplane leaves the same airport and flies due west at the speed of 1200 km/hr. How far apart will be the two planes after $1\frac{1}{2}$ hours?.	May 2022
6	A man goes 18 m due east and then 24 m due north. Find the distance of his current position from the starting point?.	Aug 2022
7	Two poles of height 'a' meters and 'b' meters are 'p' meters apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\frac{ab}{a+b}$ meters.	April 2023
8	Show that the angle bisectors of a triangle are concurrent.	April 2026
9	ABC is a triangle with $\angle B = 90^\circ$, $BC = 3\text{cm}$ and $AB = 4\text{cm}$. D is a point on AC such that $AD = 1\text{cm}$ and E is the midpoint of AB. Join D and E and extend DE to meet CB at F. Find BF.	(PTA - 1)

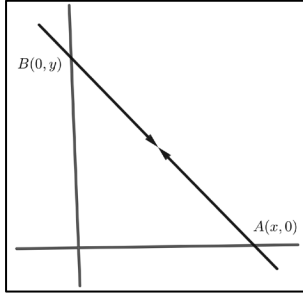
CHAPTER 5: CO-ORDINATE GEOMETRY

2 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Find the slope of a line joining the points (14, 10) and (14, -6).	Sep 2019, April 2026
2	Show that the given points are collinear $(-3, -4)$, $(7, 2)$ and $(12, 5)$	Sep 2021
3	Calculate the slope and y intercept of the straight line $8x - 7y + 6 = 0$	Sep 2021
4	Find the intercepts made by the following lines on the coordinate axes. $3x - 2y - 6 = 0$	Sep 2021
5	Show that the points $P(-1.5, 3)$, $Q(6, -2)$, $R(-3, 4)$ are collinear.	May 2022
6	Find the equation of a straight line which has slope $-\frac{5}{4}$ and passing through the point $(-1, 2)$	May 2022, June 2023
7	If area of triangle formed by vertices $A(-1, 2)$, $B(k, -2)$ and $C(7, 4)$ is 22 sq. units. find the value of 'k'	Aug 2022
8	The line p passes through the points (3, -2), (12, 4) and the line q passes through the points (6, -2) and (12, 2). Is p parallel to q?	Aug 2022, May 2022, June 2023

S.No	PUBLIC QUESTIONS	YEAR
9	Find the slope of a line joining the given points $(5, \sqrt{5})$ with the origin	Aug 2022, April 2024
10	A cat is located at the point $(-6, -4)$ in xy plane. A bottle of milk is kept at $(5, 11)$. The cat wishes to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk.	April 2023
11	If the straight lines $12y = -(p + 3)x + 12$, $12x - 7y = 16$ are perpendicular then find 'p'	April 2023
12	Find the equation of a straight line which is parallel to the line $3x - 7y = 12$ and passing through the point $(6, 4)$.	June 2023
13	If the points $A(-3, 9)$, $B(a, b)$ and $C(4, -5)$ are collinear and if $a + b = 1$, then find a and b .	April 2024
14	Find the slope of the straight line $6x + 8y + 7 = 0$.	July 2024
15	The line through the points $(-2, a)$ and $(9, 3)$ has slope $-\frac{1}{2}$. Find the value of 'a'.	April 2025
16	Show that the straight lines $x - 2y + 3 = 0$ and $6x + 3y + 8 = 0$ are perpendicular.	April 2025
17	Find the slope of a line joining the points $(-6, 1)$ and $(-3, 2)$.	July 2025
18	Find the intercepts made by the line $4x + 3y + 12 = 0$ on the coordinate axes.	July 2025
19	Find the equation of a straight line whose intercepts on the x and y axes are 4 and -6 respectively.	April 2026
20	Find the area of the triangle whose vertices are $(-3, 5)$, $(5, 6)$ and $(5, -2)$	July 2026
21	Calculate the slope and y intercept of the straight line $8x - 7y + 6 = 0$	July 2026
22	The vertices of a triangle are $A(-1, 3)$, $B(1, -1)$ and $C(5, 1)$. Find the length of the median through the vertex C.	(DMQ)
23	Find the equation of a line passing through the point $(-4, 3)$ and having slope $-\frac{7}{5}$.	(PTA-1)
24	Show that the straight lines $3x - 5y + 7 = 0$ and $15x + 9y + 4 = 0$ are perpendicular.	(PTA-3)

5 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	A straight line AB cuts the co-ordinate axes at A and B. If the mid-point of AB is $(2, 3)$, find the equation of AB. 	Sep 2019
2	Find the value of ' k ' if the area of quadrilateral is 28 sq. units, whose vertices are $(-4, -2)$, $(-3, k)$, $(3, -2)$ and $(2, 3)$	Sep 2019, April 2025
3	Find the equation of the median and altitude of $\triangle ABC$ through A where the vertices are $A(6,2)$, $B(-5, -1)$ and $C(1,9)$	Sep 2021 July 2026
4	If the points $P(-1, -4)$, $Q(b, c)$ and $R(5, -1)$ are collinear and if $2b + c = 4$, then find the values of ' b ' and ' c '	Sep 2021
5	Find the equation of the line passing through $(22, -6)$ and having intercept on X-axis exceeds the intercept on Y-axis by 5 units.	May 2022
6	A quadrilateral has vertices $A(-4, -2)$, $B(5, -1)$, $C(6,5)$ and $D(-7,6)$. Show that the mid-points of its sides form a parallelogram.	May 2022
7	A cat is located at the point $(-6, -4)$ in xy plane. A bottle of milk is kept at $(5, 11)$. The cat wishes to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk.	Aug 2022
8	Find the area of the quadrilateral formed by the points $(8,6)$, $(5,11)$, $(-5,12)$ and $(-4,3)$.	Aug 2022, April 2023, April 2026
9	Find the equation of a straight line parallel to X-axis and passing through the point of intersection of the lines $7x - 3y = -12$ and $2y = x + 3$	April 2023
10	Find the area of the quadrilateral whose vertices are at $(-9, -2)$, $(-8, -4)$, $(2,2)$ and $(1, -3)$	June 2023, April 2024, July 2025
11	Find the equation of the perpendicular bisector of the line joining the points $A(-4,2)$ and $B(6, -4)$.	June 2023, April 2024
12	Find the area of the quadrilateral whose vertices are at $(-9,0)$, $(-8,6)$, $(-1, -2)$ and $(-6, -3)$.	July 2024
13	Find the equation of a straight line through the intersection of lines $7x + 3y = 10$, $5x - 4y = 1$ and parallel to the line $13x + 5y + 12 = 0$	July 2024
14	A line makes positive intercepts on coordinate axes whose sum is 7 and it passes through $(-3,8)$. Find its equation.	April 2025

S.No	PUBLIC QUESTIONS	YEAR
15	Find the equation of a straight line perpendicular to $2x - 3y + 6 = 0$ and passes through the midpoint of the line segment joining the points $(-4, -7)$ and $(6, 5)$.	July 2025
16	$A(-3,0)$, $B(10, -2)$ and $C(12,3)$ are the vertices of $\triangle ABC$. Find the equation of the altitude through A.	April 2026
17	A straight line $7x + 16y - 56 = 0$ meets the X and Y axis at A and B respectively. Find the area of $\triangle OAB$ where O is the origin.	July 2026

CHAPTER 6: TRIGONOMETRY

2 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Prove that $\sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sec\theta + \tan\theta$	Sep 2019
2	Find the angle of elevation of the top of a tower from a point on the ground, which is 30m away from the foot of a tower of height $10\sqrt{3}$ m.	Sep 2021, Aug 2022
3	From the top of a rock $50\sqrt{3}$ m height, the angle of depression of a car on the ground is observed to be 30° . Find the distance of the car from the rock.	May 2022, April 2025
4	Prove that $\frac{\sec\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta} = \cot\theta$	April 2023
5	Prove that $\tan^2\theta - \sin^2\theta = \tan^2\theta\sin^2\theta$	June 2023
6	Prove that $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \csc\theta + \cot\theta$.	April 2024, April 2026
7	Prove the following identity $\frac{1-\tan^2\theta}{\cot^2\theta-1} = \tan^2\theta$	July 2024
8	Prove $\frac{\cos\theta}{1+\sin\theta} = \sec\theta - \tan\theta$	July 2025
9	Prove that $\sqrt{\frac{1+\tan^2\theta}{1+\cot^2\theta}} = \tan\theta$	July 2026
10	Show that, $\sqrt{\frac{\sec\theta-\tan\theta}{\sec\theta+\tan\theta}} = \frac{1-\sin\theta}{\cos\theta}$.	(PTA-5)

5 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	From the top of the tower 60m high the angles of depression of the top and bottom of a vertical lamp post are observed to be 38° and 60° respectively. Find the height of the lamp post. ($\tan 38^\circ = 0.7813, \sqrt{3} = 1.732$)	Sep 2019
2	Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between the two ships. ($\sqrt{3} = 1.732$)	Sep 2021, June 2023, April 2024, April 2026
3	From a point on the ground, the angles of elevation of the bottom and top of a tower fixed at the top of a 30 m high building are 45° and 60° respectively. Find the height of the tower. ($\sqrt{3} = 1.732$)	May 2022
4	To a man standing outside his house, the angles of elevation of the top and bottom of a window are 60° and 45° respectively. If the height of the man is 180 cm and if he is 5m away from the wall, what is the height of the window?. ($\sqrt{3} = 1.732$)	Aug 2022
5	From the top of a lighthouse, the angle of depression of two ships on the opposite sides of it are observed to be 30° and 60° . If the height of the lighthouse is h meters and the line joining the ships passes through the foot of the lighthouse, show that the distance between the ships is $\frac{4h}{\sqrt{3}}$ m.	April 2023
6	Prove that $\sqrt{\frac{1+\sin\theta}{1-\sin\theta}} + \sqrt{\frac{1-\sin\theta}{1+\sin\theta}} = 2\sec\theta$	June 2023
7	A pole 5 m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point 'A' on the ground is 60° and the angle of depression to the point 'A' from the top of the tower is 45° . Find the height of the tower. ($\sqrt{3} = 1.732$)	July 2024
8	Prove that: $\frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} + \frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} = 2$.	April 2025
9	From the top of a 12 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 30° . Determine the height of the tower.	July 2025
10	An aeroplane at an altitude of 1800 m finds that two boats are sailing towards it in the same direction. The angles of depression of the boats as observed from the aeroplane are 60° and 30° respectively. Find the distance between the two boats. ($\sqrt{3} = 1.732$)	July 2026
11	A 1.2m tall girl spots a balloon moving with the wind in a horizontal line at a height of 88.2m from the ground. The angle of the balloon from the eyes of the girl at an instant is 60° . After some times the angle of elevation reduces to 30° . Find the distance travelled by the balloon during the	(PTA-2)

S.No	PUBLIC QUESTIONS	YEAR
	interval.	

CHAPTER 7: MENSURATION

2 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Find the volume of the iron used to make a hollow cylinder of height 9 cm and whose internal and external radii are 3cm and 5cm respectively.	Sep 2019
2	Find the diameter of a sphere whose surface area is 154 m^2	Sep 2019
3	If the base area of a hemispherical solid is 1386 sq. meters, then find its total surface area?	Sep 2019, April 2024, July 2025
4	The heights of two right circular cones are in the ratio 1: 2 and the perimeters of their bases are in the ratio 3: 4. Find the ratio of their volumes.	Sep 2021
5	Find the volume of a cylinder whose height is 2m and whose base area is 250 m^2	Sep 2021, April 2024
6	The radius of a spherical balloon increases from 12cm to 16cm as air being pumped into it. Find the ratio of the surface area of the balloons in the two cases.	May 2022
7	The volumes of two cones of same base radius are 3600 cm^3 and 5040 cm^3 . Find the ratio of heights.	May 2022, July 2024, April 2025
8	If the total surface area of a cone of radius 7cm is 704cm^2 , then find its slant height.	Aug 2022
9	The radius and height of a cylinder are in the ratio 5: 7 and its curved surface area is 5500 sq. cm. Find its radius and height.	Aug 2022
10	The radius of a conical tent is 7m and height is 24m. Calculate the length of the canvas used to make the tent if the width of the rectangular canvas is 4m?	April 2023
11	If the ratio of radii of two spheres is 4: 7, find the ratio of their volumes.	April 2023, April 2025
12	The curved surface area of a right circular cylinder of height 14cm is 88cm^2 Find the diameter of the cylinder.	June 2023
13	The volume of a solid right circular cone is 11088 cm^3 If its height is 24cm then find the radius of the cone.	June 2023
14	A cylindrical drum has a height of 20 cm and base radius of 14 cm. Find its curved surface Area?.	July 2024 July 2026

S.No	PUBLIC QUESTIONS	YEAR
15	The slant height of a frustum of a cone is 5 cm and the radii of its ends are 4 cm and 1 cm. Find its curved surface area.	July 2025
16	Find the ratio of the volume of a cube to that of a largest sphere which exactly fits into the cube.	April 2026
17	Find the number of spherical lead shots, each of diameter 6cm that can be made from a solid cuboids of lead having dimensions 24cm $\times$ 22cm $\times$ 12cm.	(PTA-2)

5 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	A cylindrical glass with diameter 20 cm has water to a height of 9cm. A small cylindrical metal of radius 5cm and height 4cm is immersed it completely. Calculate the rise of the water in the glass.	Sep 2019
2	If the radii of the circular ends of a frustum which is 45 cm high are 28 cm and 7cm, find the volume of the frustum.	Sep 2021, April 2024
3	A toy is in the shape of a cylinder surmounted by a hemisphere. The height of the toy is 25cm. Find the total surface area of the toy if its common diameter is 12cm.	Sep 2021, July 2025
4	A container open at the top is in the form of a frustum of a cone of height 16cm with radii of its lower and upper ends are 8cm and 20cm respectively. Find the cost of milk which can completely fill a container at the rate of 540 per liter.	May 2022
5	Nathan, an engineering student was asked to make a model shaped like a cylinder with two cones attached at its two ends. The diameter of the model is 3cm and its length 12cm. If each cone has height 2cm, find the volume of the model that Nathan made.	May 2022, April 2025
6	A cylindrical drum has height of 20cm and base radius of 14cm. Find its curved surface area and the total surface area.	Aug 2022
7	If the circumference of base of a conical wooden piece is 484 cm then find its volume when its height is 105cm.	Aug 2022 July 2026
8	The radius and height of a cylinder are in the ratio 5: 7 and its curved surface area is 5500 sq. cm. Find its radius and height.	April 2023
9	Arul has to make arrangements for the accommodation of 150 persons for his family function. For this purpose, he plans to build a tent which is in the shape of cylinder surmounted by cone. Each person occupies 4 sq. m of the space on ground and 40 cu. Meter of air to breathe. What should be the height of the conical part of the tent if the height of cylindrical part is 8m?	April 2023
10	A metallic sphere of radius 16 cm is melted and recast into small spheres each of radius 2cm. How many small spheres can be obtained?.	June 2023, July 2024, April 2025

S.No	PUBLIC QUESTIONS	YEAR
11	A right circular cylindrical container of base radius 6 cm and height 15 cm is full of ice-cream. The ice-cream is to be filled in cones of height 9 cm and base radius 3 cm, having a hemispherical cap. Find the number of cones needed to empty the container.	April 2024
12	The internal and external diameters of a hollow hemispherical vessel are 20 cm and 28 cm respectively. Find the cost to paint the vessel all over at 0.14 per cm^2 .	July 2024
13	An Aluminum sphere of radius 12 cm is melted to make a cylinder of radius 8 cm. Find the height of the cylinder.	July 2025
14	A garden roller whose length is 3m long and whose diameter is 2.8m is rolled to level a garden. How much area will it cover in 8 revolutions?	April 2026
15	A capsule is in the shape of a cylinder with two hemisphere stuck to each of its ends. If the length of the entire capsule is 12 mm and the diameter of the capsule is 3 mm, how much volume of medicine can it hold ?	July 2026
16	A cylindrical bucket of 32 cm high and with radius of base 18 cm, is filled with sand completely. This bucket is emptied on the ground and a conical heap of sand is formed. If the height of the conical heap is 24cm, find the radius and slant height of the heap.	(DMQ/PTA-4)
17	The radius of the conical tent is 7m and the height is 24m. Calculate the length of the canvas used to make the tent. If the width of the rectangular canvas is 4 m.	(PTA - 3)
18	A well of diameter 3m is dug 14m deep. The earth taken out of it has been spread evenly all around it in the shape of a circular ring of width 4m to form an embankment. Find the height of the embankment.	(PTA-5)
19	If the slant height of the frustum cone is 10cm and perimeters of its circular base are 18cm and 28cm respectively. What is the curved surface area of the frustum?	(PTA-6)

CHAPTER 8: STATISTICS AND PROBABILITY

2 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Find the range and coefficient of range of the following data 63, 89, 98, 125, 79, 108, 117, 68.	Sep 2019, Aug 2022
2	A die is rolled and a coin is tossed simultaneously. Find the probability that the die shows an odd number and the coin shows a head.	Sep 2021, June 2023, April 2026
3	Two coins are tossed together. What is the probability of getting different faces of the coins?	May 2022, July 2025

S.No	PUBLIC QUESTIONS	YEAR														
4	A bag contains 5 red balls, 6 white balls, 7 green balls, 8 black balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is (i) white ball (ii) black or red ball	April 2023														
5	A and B are two candidates seeking admission to IIT. The probability that A getting selected is 0.5 and the probability that both A and B getting selected is 0.3. Prove that the probability of B being selected is atmost 0.8.	April 2023														
6	Find the standard deviation of first 21 natural numbers.	June 2023, April 2026														
7	Find the Range and Co-efficient of range of the following data: 25,67,48,53,18,39,44.	April 2024														
8	What is the Probability that a leap year selected at random will contain 53 Saturdays?.	April 2024 July 2026														
9	A Coin is tossed thrice. What is the Probability of getting two Consecutive tails?.	July 2024														
10	Find the range of the following distribution : <table><tr><td>Age (in Years)</td><td>16 – 18</td><td>16 – 18</td><td>16 – 18</td><td>16 – 18</td><td>16 – 18</td><td>16 – 18</td></tr><tr><td>Number of Students</td><td>0</td><td>4</td><td>6</td><td>8</td><td>2</td><td>2</td></tr></table>	Age (in Years)	16 – 18	16 – 18	16 – 18	16 – 18	16 – 18	16 – 18	Number of Students	0	4	6	8	2	2	April 2025
Age (in Years)	16 – 18	16 – 18	16 – 18	16 – 18	16 – 18	16 – 18										
Number of Students	0	4	6	8	2	2										
11	A flower is selected at random from a basket containing 80 yellow, 70 red and 50 white flowers. Find the probability of selecting a yellow coloured flower.	April 2025														
12	If the range and the smallest value of a set of data are 36.8 and 13.4 respectively, then find The largest value.	July 2025														
13	The standard deviation and mean of a data are 6.5 and 12.5 respectively. Find the coefficient of variation.	July 2026														
14	The standard deviation of 20 observations is $\sqrt{6}$. If each observation is multiplied by 3, find the standard deviation and variance of the resulting observations.	(PTA - 1)														

5 Mark Questions

S.No	PUBLIC QUESTIONS	YEAR
1	The scores of a cricketer in 7 matches are 70, 80, 60, 60, 40, 90, 95. Find the standard deviation	Sep 2019
2	Two unbiased dice are rolled once. Find the probability of getting (i) a doublet (ii) the product as a prime number (iii) the sum as a prime number (iv) the sum as 1	Sep 2019, Aug 2022, Apr 2023
3	Two dice are rolled. Find the probability that sum of outcomes is (i) equal to 4, (ii) greater than 10, (iii) less than 13	Sep 2021, July 2024

S.No	PUBLIC QUESTIONS	YEAR										
4	In a class of 50 students, 28 opted for NCC, 30 opted for NSS and 18 opted both NCC and NSS. One of the students is selected at random. Find the probability that (i) The student opted for NCC but not NSS. (ii) The student opted for NSS but not NCC. (iii) The student opted for exactly one of them.	May 2022										
5	Two dice are rolled once. Find the probability of getting an even number on the first die or a total of face sum 8.	June 2023, April 2024										
6	A teacher asked the students to complete 60 pages of a record note book. Eight students have completed only 32, 35, 37, 30, 33, 36, 35 and 37 pages. Find the standard deviation of the pages completed by them.	July 2024										
7	Find the coefficient of variation 24, 26, 33,37, 29, 31.	April 2024, April 2026										
8	The total marks scored by two students, Sathya and Vidhya in 5 subjects are 460 and 480 with standard deviation 4.6 and 2.4 respectively. Who is more consistent in performance.	April 2025										
9	A bag contains 5 red balls, 6 white balls,7 green balls, 8 black balls. One ball is random from the bag. Find the probability that the ball drawn is (i) white (ii) black or red (iii) not white (iv) neither white nor black	April 2025										
10	The marks scored by the students in a slip test are given below. <table border="1"><tr><td><i>x</i></td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td><i>f</i></td><td>7</td><td>3</td><td>5</td><td>9</td></tr></table> Find the standard deviation of their marks.	<i>x</i>	4	6	8	10	<i>f</i>	7	3	5	9	July 2025
<i>x</i>	4	6	8	10								
<i>f</i>	7	3	5	9								
11	Two dice are rolled together. Find the Probability of getting a doublet or sum of faces as 4.	April 2026										
12	Find the mean and variance of the first 'n' natural numbers.	July 2026										
13	Three fair coins are tossed together. Find the probability of getting: (i) all heads											

8 MARK QUESTIONS: GEOMETRY & GRAPH

Geometry Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Construct a triangle similar to a given triangle ABC with its sides equal to $\frac{6}{5}$ of the corresponding sides of the triangle ABC (scale factor $\frac{6}{5} > 1$)	Sep 2019
2	Draw the two tangents from a point which is 10cm away from the centre of a circle of radius 5cm. Also, measure the lengths of the tangents.	Sep 2019, June 2023, July 2024
3	Construct a $\triangle PQR$ which the base $PQ = 4.5$ cm, $\angle R = 35^\circ$ and the median from R to RG is 6cm.	Sep 2021, April 2024 July 2026
4	Draw a circle of diameter 6cm from a point P, which 8cm away from its centre. Draw two tangents PA and PB to the circle and measure their lengths.	Sep 2021, Aug 2022, April 2024, April 2026
5	Construct a $\triangle ABC$ such that $AB = 5.5$ cm, $\angle C = 25^\circ$ and the altitude from C to AB is 4cm.	May 2022
6	Draw two tangents from a point which is 5cm away from the center of a circle of diameter 6cm. Also, measure the lengths of the tangents.	May 2022
7	Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{7}{3}$ of the corresponding sides of the triangle PQR (scale factor $\frac{7}{3} > 1$)	Aug 2022, July 2024
8	Take a point which is 11 cm away from the center of a circle of radius 4cm and draw the two tangents to the circle from the point.	April 2023, April 2025
9	Draw a triangle ABC of base $BC = 8$ cm, $\angle A = 60^\circ$ and the bisector of $\angle A$ meets BC at D such that $BD = 6$ cm.	April 2023, April 2025
10	Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle PQR (scale factor $\frac{3}{5} < 1$)	June 2023, April 2026
11	Construct a $\triangle PQR$ in which $QR = 5$ cm, $\angle P = 40^\circ$ and the Median PG from P to QR is 4.4 cm. Find the length of the Altitude from P to QR.	July 2024
12	Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{7}{4}$ of the corresponding sides of the triangle PQR. (scale factor $\frac{7}{4} > 1$)	July 2025
13	Draw a tangent to the circle from the point P having radius 3.6 cm, and centre at O, point P is at a distance 7.2 cm from the centre.	July 2026

Graph Questions

S.No	PUBLIC QUESTIONS	YEAR
1	Graph the following quadratic equation $x^2 - 8x + 16 = 0$ and state their nature of solution.	Sep 2019

S.No	PUBLIC QUESTIONS	YEAR												
2	Draw the graph of $y = 2x^2 - 3x - 5$ and hence use it to solve $2x^2 - 4x - 6 = 0$	Sep 2019, April 2024												
3	Draw the graph of $x^2 + x - 12 = 0$ and state the nature of their solution.	Sep 2021, July 2025												
4	Draw the graph of $y = x^2 + 3x - 4$ and hence use it to solve $x^2 + 3x - 4 = 0$	Sep 2021 July 2026												
5	Draw the graph of $x^2 - 4x + 4 = 0$ and state the nature of their solution	May 2022												
6	Draw the graph of $y = x^2 - 4x + 3$ and use it solve $x^2 - 6x + 9 = 0$	May 2022, Aug 2022												
7	Graph the following quadratic equation $x^2 - 9x + 20 = 0$ and state their nature of solution.	Aug 2022, April 2025												
8	Varshika drew 6 circles with different sizes. Draw a graph for the relationship between the diameter and circumference of each circle as shown in the table and use it to find the circumference of a circle when its diameter is 6 cm. <table><tr><td>Diameter (x)cm</td><td>1</td><td>23</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Circumference (y) cm</td><td>3.1</td><td>6.2</td><td>9.3</td><td>12.4</td><td>15.5</td></tr></table>	Diameter (x)cm	1	23	3	4	5	Circumference (y) cm	3.1	6.2	9.3	12.4	15.5	April 2023
Diameter (x)cm	1	23	3	4	5									
Circumference (y) cm	3.1	6.2	9.3	12.4	15.5									
9	Draw the graph of $y = x^2 - 5x - 6$ and hence use it to solve $x^2 - 5x - 14 = 0$	April 2023												
10	Graph the following linear function $y = \frac{1}{2}x$. Identify the constant of variation and verify it with the graph. Also (i) find y when $x = 9$ (ii) find x when $y = 7.5$	June 2023, April 2026												
11	Draw the graph of $y = x^2 - 4$ and hence solve $x^2 - x - 12 = 0$	June 2023												
12	Draw the graph of $xy = 24$, $x, y > 0$ Using the graph find, (i) y when $x = 3$ and (ii) x when $y = 6$.	April 2024												
13	Draw the graph of $y = x^2 + x - 2$ and hence solve $x^2 + x - 2 = 0$.	July 2024												
14	The following table shows the data about the number of pipes and the time taken to fill the same tank. <table><tr><td>No. of. pipes x</td><td>2</td><td>3</td><td>6</td><td>9</td></tr><tr><td>Time Taken y (in mins)</td><td>45</td><td>30</td><td>15</td><td>10</td></tr></table> Draw the graph for the above data and hence (i) Find the time taken to fill the tank when five pipes are used. (ii) Find the number of pipes when the time is 9 minutes.	No. of. pipes x	2	3	6	9	Time Taken y (in mins)	45	30	15	10	July 2024		
No. of. pipes x	2	3	6	9										
Time Taken y (in mins)	45	30	15	10										

S.No	PUBLIC QUESTIONS	YEAR												
15	A bus is travelling at a uniform speed of 50 km/hr. Draw the time -distance graph and hence find. (i) the constant of variation. (ii) how far will it travel in 90 minutes? (iii) The time required to cover a distance of 300 km.	April 2025												
16	A school announces that for a certain competition the cash prize will be distributed for all the participants equally as shown below: <table><tr><td>No .of Participants x</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>Amount for each participant y in ₹</td><td>180</td><td>90</td><td>60</td><td>45</td><td>36</td></tr></table> (i) Find the constant of variation. (ii) Graph the above data and hence find how much each participant will get if the number of participants are 12.	No .of Participants x	2	4	6	8	10	Amount for each participant y in ₹	180	90	60	45	36	July 2025
No .of Participants x	2	4	6	8	10									
Amount for each participant y in ₹	180	90	60	45	36									
17	Graph the quadratic equation $x^2 - 6x + 9 = 0$ and state its nature of solutions.	April 2026												
18	A garment shop announces a flat 50% discount on every purchase of items for their customers. Draw the graph for the relation between the Marked price and the Discount. Hence find: (i) the marked price when a customer gets a discount of ₹ 3,250 (from graph). (ii) the discount when the marked price is ₹ 2,500.	July 2026												

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